Investigating the Causes of Rusting

Aim: To investigate factors which affect rusting.

Equipment:

- Safety goggles
- 4 test tubes
- Small beaker
- Stirring rod
- Test tube rack
- Rubber bung for test tube
- Cotton wool
- Forceps
- Marker pen
- 4 iron nails
- Anhydrous calcium chloride granules
- Vegetable oil
- Dropping pipette
- Deionised water
- Boiled deionised water
- Table salt



Safety Information

Calcium chloride - causes irritation to skin, eyes and respiratory organs. Harmful if inhaled or swallowed.

Wear goggles at all times.

Glassware – risk of cuts if broken.

Apply pressure to wounds and inform teacher immediately.

Method:

Step 1: Label the tests tube 1, 2, 3 and 4

Step 2: In test tube 1, fill $\frac{1}{4}$ of the test tube with deionised water and add a nail

Step 3: In test tube 2, fill $\frac{1}{4}$ of the test tube with boiled deionised water and add a nail. Carefully drop some oil onto the water – with the pipette – to create an air tight layer

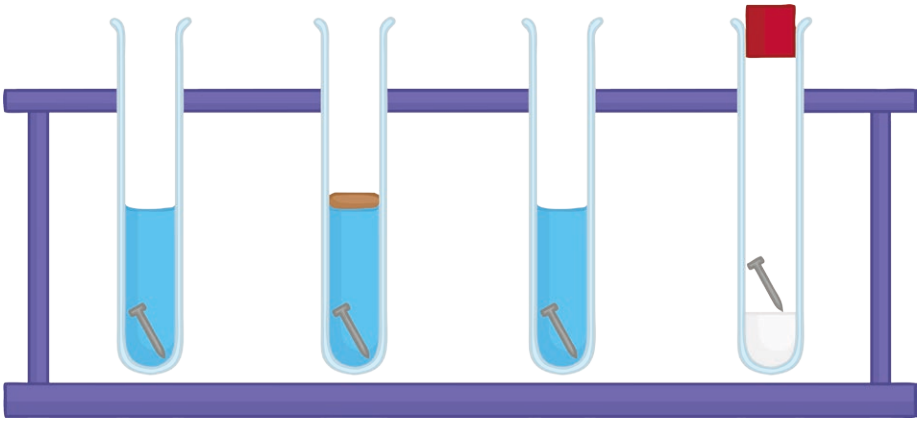
Step 4: Mix salt into some deionised water in the beaker, and stir with the rod until dissolved

Step 5: In test tube 3, fill $\frac{1}{4}$ of the test tube with the salt solution you just made, and add a nail

Step 6: In test tube 4, add about 2 cm depth of calcium chloride granules and add a nail. Seal the test tube with the rubber bung

Step 7: Leave the nails for 3 days

Step 8: Record your observations into a suitable table



Test tube conditions:

Test tube 1 – water and air

Test tube 2 – water, no air (removed during boiling process / sealed by oil layer)

Test tube 3 – water, air and salt

Test tube 4 – no water, air (calcium chloride absorbs the water, bung prevents moisture from air)

Questions:

1. Use the information from the box above to make a clear table of the conditions of each test tube.
2. Describe what your results have shown.
3. Explain your results using your understanding of the oxidation reaction which causes rusting.
4. Was your investigation accurate? How can you improve it?

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Answers:

1. Use the information from the box above to make a clear table of the conditions of each test tube.

Test Tube	Air	Water	Salt
1	Y	Y	N
2	N	Y	N
3	Y	Y	Y
4	Y	N	N

2. Describe what your results have shown.
   The results show that rusting occurs in the presence of water and air (test tubes 1 and 3). It should also show that salt increases the rate of rusting.

3. Explain your results using your understanding of the oxidation reaction which causes rusting.
   The oxidation reaction of iron requires oxygen, which is in the air, and dissolved in the water. Although water was present in test tube 2, it had been boiled so any dissolved oxygen had been removed. Test tube 4 did not show any rusting because only air was present.

4. Was your investigation accurate? How can you improve it?
   The accuracy of the measurements of the different chemicals used was not very accurate. Rather than measuring by cm depth it would be more accurate to measure the mass in g, using a top pan balance. Measuring the liquids as $\frac{1}{4}$ filled test tubes was also not accurate and could be improved by measuring in millilitres using a measuring cylinder.